

Hemanshu Boppana

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EDUCATION

University of Florida

Gainesville, FL

B.S. in Computer Science, Minor in Statistics; GPA: 3.86/4.00

Expected Graduation: May 2027

Relevant Coursework: Data Structures & Algorithms, Operating Systems, Computer Architecture, Information & Database Systems, Natural Language Processing, Math in Machine Learning

Awards: SwampHacks 2026 Winner | SASEHacks 2024 Winner | Azure DP-900 Certified

EXPERIENCE

Software Engineer Intern

Jan 2026 – Present

Diploy

Los Angeles, CA

- Shipped an AI-powered resume builder adopted across **10 university pilot programs**, scaling to **1,000+ active users** and **250+ generated resumes**. Designed backend services integrating the **Anthropic API** to process **3K-token requests** under low-latency constraints.
- Wrote data pipelines in **TypeScript** and **PowerShell** to ingest and normalize **4.5M labor-market records** across 50 states, enabling reliable data access for analytics and reporting. Built integration tests covering **SSE streaming** and API contract correctness, ensuring data quality and system stability.

Machine Learning Researcher

Mar 2025 – Present

Intelligent Clinical Care Center (iHeal Lab) @ University of Florida

Gainesville, FL

- Shaped model architecture selection and **evaluation methodology** for the lab's heart failure and delirium prediction research by **synthesizing 500+ publications** on clinical machine learning and ICU outcome prediction.
- Built initial evaluation pipelines on **HiPerGator distributed GPU clusters** for a heart failure classifier achieving **90% precision**, establishing baseline metrics and automated validation workflows for ongoing development.

Research Engineer

Feb 2025 – Sep 2025

Artificial Intelligence Research (AIR) Lab @ University of Florida

Gainesville, FL

- Built data-ingestion pipelines using **Selenium** and **BeautifulSoup** to collect and structure **10,000+ records** from political media sources for downstream misinformation-detection models with automated validation.
- Created reusable **ETL workflows** in Python and Pandas on **HiPerGator**, cutting data processing runtime by **60%** and improving pipeline throughput by **25%**.

PROJECTS

VectorDB | C++, OpenMP, AVX2, HNSW, SIMD

Feb 2026 – Mar 2026

- Engineered a vector database from scratch in C++ for semantic search, achieving **91% Recall@10** on 50K-vector nearest-neighbor queries in **3.8 ms** using HNSW indexing.
- Applied AVX2 SIMD intrinsics and OpenMP parallelism to improve throughput **3.75x** and cut index build time from 611.8s to 4.6s (**132x speedup**).

Explainable | TypeScript, React, Flask, Gemini API, ElevenLabs, Twilio, Redis/Upstash

Jan 2026

- Built an AI tool that explains complex academic topics over web and phone by coordinating **5 microservices** (speech synthesis, LLM inference, telephony). Won **Best Use of Gemini API** at SwampHacks XI.
- Reduced end-to-end latency by **30%+** with **Redis** caching and service-layer optimizations. Designed fault-tolerant **Twilio** integrations for low-bandwidth environments.

Personal Tutoring Platform | TypeScript, Next.js, Supabase, AWS (S3, EC2, Lambda), Stripe, Vercel

Jun 2025

- Launched a tutoring marketplace generating **\$800+ monthly recurring revenue**, owning the full product lifecycle from design through deployment and acquiring paying users organically.
- Integrated **Stripe payments** and Cal.com scheduling via **REST APIs**. Used **AWS Lambda** for serverless compute and **S3** for storage, with **CI/CD pipelines** on **Vercel**, reducing deploy time by **90%**.

TECHNICAL SKILLS

Languages: Python, C++, Java, TypeScript, JavaScript, SQL

Frameworks: React, Next.js, Node.js, FastAPI, Flask, REST APIs, PyTorch, Pandas

Cloud & DevOps: AWS (S3, EC2, Lambda), Azure, GCP, Docker, Git, CI/CD, Linux

Data & Infrastructure: PostgreSQL, MongoDB, Redis, Supabase, Microservices, OpenMP, SIMD

Tools & APIs: GenAI/LLM APIs (Anthropic, Gemini), Stripe, Twilio, Vercel, HiPerGator (HPC)